

A Crash Course on Graphs

Foundations for Graph Learning and Graph Neural Networks

Lecture Notes — 90 minutes

Abstract

These notes are a self-contained, fast-paced introduction to graph theory aimed at students who will move on to graph representation learning and Graph Neural Networks (GNNs). We cover the basic vocabulary (directed/undirected, weighted graphs, walks, paths, connectivity), local and global numerical descriptors (degree, degree distribution, clustering, centrality), and the core matrix representations: adjacency, incidence, transition, and (normalized) Laplacian. We then study the spectrum of the Laplacian and its meaning in terms of connectivity and smoothness, which directly motivates spectral graph convolutions and the message-passing paradigm of GNNs.

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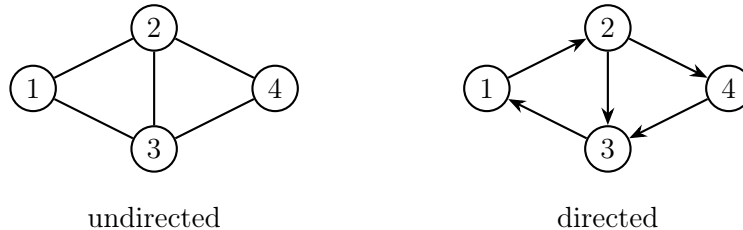
1 What is a graph and why do we care?

A graph is the simplest mathematical object able to encode *relations* between entities. Almost every dataset that is not a regular grid (image) or a sequence (text, time series) is more naturally modelled as a graph: molecules, social networks, citation networks, knowledge graphs, road networks, meshes, point clouds, brain connectomes, recommender systems.

Classical machine learning assumes data points $x_i \in \mathbb{R}^d$ are i.i.d. samples from a distribution. On graphs this assumption fails: data points are *linked*, and the link structure carries information. Graph learning is the program of designing models that exploit this structure.

Definition 1.1 (Graph). A graph is a pair $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ is a finite set of vertices (or nodes) and $\mathcal{E}$ is a set of edges. We write $n = |\mathcal{V}|$ and $m = |\mathcal{E}|$.

- If edges are unordered pairs $\{u, v\}$, the graph is undirected.
- If edges are ordered pairs (u, v) , the graph is directed (a digraph); u is the tail, v the head.
- If each edge e carries a real number $w(e)$, the graph is weighted.
- An edge (v, v) is a self-loop; multiple edges between the same pair give a multigraph. A graph without self-loops or multiple edges is simple.



Definition 1.2 (Walks, paths, cycles). A walk of length k from u to v is a sequence $u = v_0, v_1, \dots, v_k = v$ such that $\{v_{i-1}, v_i\} \in \mathcal{E}$ for every i . A path is a walk with distinct vertices. A cycle is a walk with $v_0 = v_k$ and otherwise distinct vertices.

Definition 1.3 (Connectivity). An undirected graph is connected if there is a path between any two vertices. Maximal connected subgraphs are called connected components. A directed graph is strongly connected if for every (u, v) there is a directed path $u \rightarrow v$ and $v \rightarrow u$; it is weakly connected if its underlying undirected graph is connected.

Definition 1.4 (Special graphs). Some graphs reappear constantly: the complete graph K_n (every pair joined), the path P_n , the cycle C_n , the star S_n (one centre joined to $n - 1$ leaves), the grid, the bipartite graph $K_{m,n}$, regular trees, and d -regular graphs (every vertex has degree d).

2 Numerology: local and global descriptors

2.1 Degree

Definition 2.1 (Degree). For an undirected simple graph, the degree of v is $\deg(v) = |\{u : \{u, v\} \in \mathcal{E}\}|$. For a digraph, the in-degree $\deg^-(v)$ counts incoming edges, the out-degree $\deg^+(v)$ outgoing edges. For a weighted graph one analogously defines the weighted degree $\deg(v) = \sum_u w(\{u, v\})$.

Proposition 2.2 (Handshake lemma). In every undirected graph,

$$\sum_{v \in \mathcal{V}} \deg(v) = 2m.$$

In a digraph, $\sum_v \deg^+(v) = \sum_v \deg^-(v) = m$.

Proof sketch. Each edge contributes to the degree of exactly two vertices (or one tail and one head, in the directed case). $\square$

Corollary 2.3. *The number of vertices with odd degree is even.*

Degree distribution. The empirical distribution $p(k) = |\{v : \deg(v) = k\}|/n$ is a first global descriptor. Many real networks (web, citations, social) display a heavy-tailed or roughly power-law $p(k) \propto k^{-\gamma}$ degree distribution; this contrasts sharply with Erdős–Rényi random graphs $G(n, p)$, whose degree distribution is binomial.

2.2 Distances and diameter

The *geodesic distance* $d(u, v)$ is the length of a shortest path between u and v ($+\infty$ if none exists). The *eccentricity* of v is $\varepsilon(v) = \max_u d(u, v)$. The *diameter* is $\text{diam}(\mathcal{G}) = \max_v \varepsilon(v)$ and the *radius* is $\min_v \varepsilon(v)$.

2.3 Clustering and triangles

Definition 2.4 (Local clustering coefficient). *For a vertex v with $\deg(v) \geq 2$,*

$$C(v) = \frac{\#\{\text{edges among neighbours of } v\}}{\binom{\deg(v)}{2}} \in [0, 1].$$

The global clustering coefficient is $C(\mathcal{G}) = \frac{1}{n} \sum_v C(v)$.

A high clustering coefficient indicates that “a friend of a friend is also a friend”, which is one of the empirical hallmarks of social networks (*small-world* property).

2.4 Centrality measures

There are many ways to formalise “how important is a node”:

- **Degree centrality:** $\deg(v)/(n-1)$.
- **Closeness centrality:** $(\sum_u d(v, u))^{-1}$.
- **Betweenness centrality:** fraction of shortest paths in $\mathcal{G}$ that pass through v .
- **Eigenvector / PageRank centrality:** a node is important if its neighbours are; we will see that this is naturally expressed via the adjacency matrix.

3 Matrix representations of a graph

Order the vertices $\mathcal{V} = \{v_1, \dots, v_n\}$ and the edges $\mathcal{E} = \{e_1, \dots, e_m\}$ once and for all. Every interesting matrix below depends on this ordering up to a permutation conjugation; spectral quantities will be permutation invariants.

3.1 Adjacency matrix

Definition 3.1. *The adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is defined by*

$$\mathbf{A}_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in \mathcal{E} \text{ (or } (v_i, v_j) \in \mathcal{E} \text{ for digraphs),} \\ 0 & \text{otherwise.} \end{cases}$$

For a weighted graph $\mathbf{A}_{ij} = w(\{v_i, v_j\})$ (or 0 if not adjacent).

- For undirected graphs, $\mathbf{A} = \mathbf{A}^\top$.
- Row sums recover the degree: $\mathbf{D} = \text{diag}(\mathbf{A} \mathbf{1})$.
- For a digraph, $\mathbf{A}$ is generally non-symmetric; row sums give out-degrees, column sums in-degrees.

Proposition 3.2 (Walk counting). *For any $k \in \mathbb{N}$, $(\mathbf{A}^k)_{ij}$ equals the number of walks of length k from v_i to v_j .*

Proof sketch. Induction on k : $(\mathbf{A}^{k+1})_{ij} = \sum_\ell (\mathbf{A}^k)_{i\ell} \mathbf{A}_{\ell j}$, which sums walks of length k to v_ℓ extended by an edge $\ell \rightarrow j$. $\square$

Example 3.3 (Triangle counting). The number of triangles in an undirected simple graph is $\frac{1}{6} \text{tr}(\mathbf{A}^3)$: each triangle contributes $3! = 6$ closed walks of length 3.

3.2 Degree matrix

$$\mathbf{D} = \text{diag}(\deg(v_1), \dots, \deg(v_n)) \in \mathbb{R}^{n \times n}.$$

It is diagonal, positive semi-definite, and the bookkeeping device that pairs with $\mathbf{A}$.

3.3 Incidence matrix

Definition 3.4 (Undirected case). *Fix an arbitrary orientation of each edge. The (signed) incidence matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ has columns indexed by edges:*

$$\mathbf{B}_{ie} = \begin{cases} +1 & v_i \text{ is the head of } e, \\ -1 & v_i \text{ is the tail of } e, \\ 0 & \text{otherwise.} \end{cases}$$

The unsigned incidence matrix replaces -1 with $+1$. The signed version is the discrete analogue of the gradient operator, mapping a function on edges to a function on vertices.

Proposition 3.5. *For an undirected graph with the signed incidence matrix above,*

$$\mathbf{B} \mathbf{B}^\top = \mathbf{L} = \mathbf{D} - \mathbf{A}.$$

Proof sketch. $(\mathbf{B} \mathbf{B}^\top)_{ij} = \sum_e \mathbf{B}_{ie} \mathbf{B}_{je}$. For $i = j$ each incident edge contributes 1, giving $\deg(v_i)$. For $i \neq j$, only the (at most one) edge between them contributes, with sign -1 , hence $-\mathbf{A}_{ij}$. $\square$

3.4 Random-walk matrix

Definition 3.6. *Assume $\deg(v_i) > 0$ for all i . The transition matrix of the simple random walk is*

$$\mathbf{P} = \mathbf{D}^{-1} \mathbf{A}, \quad P_{ij} = \frac{\mathbf{A}_{ij}}{\deg(v_i)}.$$

$\mathbf{P}$ is row-stochastic; iterating $\mathbf{p}_{t+1} = \mathbf{p}_t \mathbf{P}$ describes the evolution of a probability distribution under one step of the random walk. PageRank is a small modification of this matrix.

4 The graph Laplacian

The graph Laplacian is the central object linking combinatorics, geometry and learning on graphs.

4.1 Definitions

Definition 4.1 (Combinatorial / unnormalized Laplacian). *For an undirected (possibly weighted) graph,*

$$\mathbf{L} = \mathbf{D} - \mathbf{A}.$$

Definition 4.2 (Normalized Laplacians). *Assume no isolated vertices. The symmetric normalized Laplacian is*

$$\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2},$$

and the random-walk Laplacian is

$$\mathbf{L}_{\text{rw}} = \mathbf{I} - \mathbf{D}^{-1} \mathbf{A} = \mathbf{I} - \mathbf{P}.$$

The two normalized variants are similar: $\mathbf{L}_{\text{rw}} = \mathbf{D}^{-1/2} \mathbf{L}_{\text{sym}} \mathbf{D}^{1/2}$, hence they share the same spectrum.

4.2 Why is it called a Laplacian?

For $f : \mathcal{V} \rightarrow \mathbb{R}$ identified with a vector $\mathbf{f} \in \mathbb{R}^n$, a direct computation gives

$$(\mathbf{L}\mathbf{f})_i = \sum_{j:\{i,j\} \in \mathcal{E}} (f(v_i) - f(v_j)),$$

i.e. at each vertex $\mathbf{L}$ accumulates the differences with the neighbours. This is the discrete analogue of $-\Delta f = -\text{div}(\nabla f)$ on a Riemannian manifold. Concretely, on a regular grid graph, $\mathbf{L}$ is exactly the (negative) finite-difference Laplacian up to boundary effects.

4.3 Quadratic form and the Dirichlet energy

Proposition 4.3 (Dirichlet energy). *For every $\mathbf{f} \in \mathbb{R}^n$,*

$$\mathbf{f}^\top \mathbf{L} \mathbf{f} = \sum_{\{i,j\} \in \mathcal{E}} w_{ij} (f(v_i) - f(v_j))^2 \geq 0.$$

Proof. Use $\mathbf{L} = \mathbf{B}\mathbf{B}^\top$ from Proposition 3.5 (weighted version: scale columns of $\mathbf{B}$ by $\sqrt{w_e}$): then $\mathbf{f}^\top \mathbf{L} \mathbf{f} = \|\mathbf{B}^\top \mathbf{f}\|^2$, which expands to the announced sum. $\square$

Interpretation. $\mathbf{f}^\top \mathbf{L} \mathbf{f}$ is small precisely when $\mathbf{f}$ varies smoothly along edges. This is the variational principle behind Laplacian-based regularization, label propagation, and (as we will see) spectral GNNs.

4.4 Spectrum of the Laplacian

Because $\mathbf{L}$ is symmetric and positive semi-definite, it admits a real eigendecomposition

$$\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n,$$

with $\mathbf{U}$ orthogonal and columns $\mathbf{u}_1, \dots, \mathbf{u}_n$ orthonormal eigenvectors.

Theorem 4.4 (Algebraic connectivity). *For an undirected graph with n vertices:*

- (i) $\lambda_1 = 0$, with eigenvector $\mathbf{1} = (1, \dots, 1)^\top$.
- (ii) The multiplicity of the eigenvalue 0 equals the number of connected components of $\mathcal{G}$.
- (iii) Equivalently, $\mathcal{G}$ is connected $\iff \lambda_2 > 0$. The value λ_2 is called the algebraic connectivity or Fiedler value.

Proof idea. $\mathbf{L}\mathbf{1} = 0$ since rows of $\mathbf{L}$ sum to zero. The Dirichlet energy gives $\mathbf{L}\mathbf{f} = 0 \iff f$ is constant on every connected component, so the kernel has dimension equal to the number of components. $\square$

Fiedler vector and spectral clustering. The eigenvector $\mathbf{u}_2$ associated with λ_2 (the *Fiedler vector*) provides a one-dimensional embedding of the graph and is a classical heuristic for graph bipartitioning: one cuts on the sign of its entries. More generally, the first k non-trivial eigenvectors of $\mathbf{L}$ (or $\mathbf{L}_{\text{sym}}$) furnish the embedding used by *spectral clustering*.

Bounds. For the symmetric normalized Laplacian, $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n \leq 2$, with $\lambda_n = 2$ iff a connected component is bipartite.

Example 4.5 (Path P_4). For $\mathcal{V} = \{1, 2, 3, 4\}$ with edges $\{12, 23, 34\}$,

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{D} = \text{diag}(1, 2, 2, 1), \quad \mathbf{L} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

The eigenvalues are $\{0, 2 - \sqrt{2}, 2, 2 + \sqrt{2}\}$. The connected graph has a one-dimensional kernel, as predicted.

5 From graphs to graph learning

A typical graph-learning setup adds *node features* $\mathbf{X} \in \mathbb{R}^{n \times d}$ (one row per vertex) and possibly edge features. The tasks are:

- **Node-level:** classify or regress on each node (e.g. which paper is in which research area).
- **Edge-level:** predict the existence or attributes of an edge (link prediction, recommendation).
- **Graph-level:** predict a property of the whole graph (molecular toxicity, scene classification).

5.1 Anomaly detection on graphs (a teaser)

Because edges express expected interactions, deviations from the typical pattern of links or features are natural anomalies: unusual subgraphs, abnormal communication patterns, fraudulent transactions in payment networks. Techniques range from structural rules (high-degree but low-clustering nodes), to spectral methods (anomalous components in the Laplacian eigenbasis), to GNN-based reconstruction errors.

5.2 Spectral and spatial views of graph convolution

A signal $\mathbf{x} \in \mathbb{R}^n$ on the vertices admits a *graph Fourier transform*

$$\hat{\mathbf{x}} = \mathbf{U}^\top \mathbf{x},$$

in the eigenbasis of $\mathbf{L}$; small eigenvalues correspond to “low frequencies” (smooth signals), large ones to “high frequencies”. A spectral filter $g(\mathbf{L})$ acts as

$$g(\mathbf{L}) \mathbf{x} = \mathbf{U} g(\boldsymbol{\Lambda}) \mathbf{U}^\top \mathbf{x}.$$

Choosing g as a low-order polynomial yields a *localized* filter (depends only on K -hop neighbourhoods) and avoids diagonalising $\mathbf{L}$. This is the design behind ChebNet and, after a first-order simplification, the celebrated GCN of Kipf & Welling:

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}\right),$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ adds self-loops and $\tilde{\mathbf{D}}$ is the corresponding degree matrix.

5.3 The message-passing paradigm (spatial view)

Most modern GNN layers are instances of *message passing*:

$$\mathbf{h}_v^{(\ell+1)} = \phi \left(\mathbf{h}_v^{(\ell)}, \bigoplus_{u \in \mathcal{N}(v)} \psi \left(\mathbf{h}_v^{(\ell)}, \mathbf{h}_u^{(\ell)}, \mathbf{e}_{uv} \right) \right),$$

with $\bigoplus$ a permutation-invariant aggregator (sum, mean, max) and ϕ, ψ learnable. Different choices recover GCN, GraphSAGE, GIN, GAT, MPNN, etc. The role of the foundational material we have just covered is to make these choices interpretable:

- the adjacency matrix (and its normalizations) defines *who* talks to whom;
- the degree matrix appears in the normalization that prevents exploding activations on hubs;
- the Laplacian and its spectrum control *smoothness* of the learnt representations and explain phenomena like *over-smoothing* (deep GCNs converge to the kernel of $\mathbf{L}_{\text{sym}}$);
- walks and powers of $\mathbf{A}$ describe the receptive field of ℓ -layer message passing.

6 Recap and reading list

Take-aways.

1. A graph is $(\mathcal{V}, \mathcal{E})$, possibly directed or weighted; degrees, walks, distances and clustering are the basic local/global descriptors.
2. Three matrices encode almost everything: adjacency $\mathbf{A}$, degree $\mathbf{D}$, signed incidence $\mathbf{B}$. They satisfy $\mathbf{L} = \mathbf{D} - \mathbf{A} = \mathbf{B}\mathbf{B}^\top$.
3. The Laplacian is symmetric PSD; its spectrum encodes connectivity (multiplicity of 0) and smoothness (λ_2 , Fiedler vector); normalized variants have spectrum in $[0, 2]$.
4. Spectral GNNs are polynomial filters in $\mathbf{L}$; spatial GNNs are message-passing schemes whose receptive field corresponds to powers of $\mathbf{A}$.

Suggested next steps. Spectral graph theory (Chung), random walks and mixing times, graph signal processing (Shuman et al.), and the GNN literature: GCN, GraphSAGE, GAT, GIN, and surveys on geometric deep learning.